

and scalable process.

Researchers first put a layer of graphene over a silicon wafer. The graphene-covered silicon wafer was then placed into cryogenic ammonia that contained a small amount of lithium. This process adds hydrogen to the wafer, which makes it ferromagnetic. While that is a pretty neat feat, what surprised the researchers was just how evenly the magnetism was spread across the wafer.

The key discovery was that they were able to remove hydrogen atoms from the material using an electron beam. This maintained magnetic areas in some parts of the graphene and removed magnetism from others. This means that large areas of graphene can be patterned with the electron beam to precisely tune magnetism. Since massive patterning with commercial electron beam lithography system is possible, this technique can be readily applicable for current microelectronics fabrication.

Role in spintronics. Spintronics, where the spin of electrons is used to encode information rather than charge, has long promised to be the next step in the evolution of computing. We have already seen evidence of that migration with today's disk drives that are capable of much greater storage capacities than the previous generation, due to the material phenomenon known as giant magnetoresistance.

However, it did not appear as though graphene was going to have much of an impact on the future of spintronics, because if you laid it out flat it did not seem to have any effect on the spin of electrons. This changed when it was discovered that if you put a small bend in graphene it could influence the spin of electrons.

In the latest piece of work, an electron spin has been preserved for an extended distance using large-area graphene. These results

would attract a lot of attention in the research community and put graphene on the map for applications in spintronic components.

Paves the way for harmless x-rays. Electromagnetic waves in the terahertz range make it possible to see into the body without using harmful x-rays. Radiation is moderated by water and therefore stops after about a centimetre. But it is precisely its sensitivity to fluids that makes it possible to create an image of various tissues in the body, or to see through clothing, for example. Terahertz radiation requires materials whose electrons can move extremely rapidly, and there are great hopes for graphene.

A team has already constructed a graphene transistor that works at more than ten gigahertz. This research is now proceeding into the terahertz range, which opens up entirely new possibilities. This could lead to soft x-rays for applications in biomedicine or security checks at airports, for instance, where there has been a lack of suitable component materials to deal with sufficiently high frequencies.

Another future scenario is short-distance radars that can penetrate fog and rain up to 100 metres, which might facilitate driving in bad weather.

Technological bottlenecks

The problem that prevented graphene from initially being available for developmental research in commercial uses was that the creation of high-quality graphene was a very expensive and complex process (of chemical vapour deposition). It involved the use of toxic chemicals to grow graphene as a monolayer by exposing platinum, nickel or titanium-carbide to ethylene or benzene at high temperatures.

Also, it was previously impossible to grow graphene layers on a large scale using crystalline epitaxy on anything other than a metallic

substrate. This severely limited its use in electronics as it was difficult, at that time, to separate graphene layers from its metallic substrate without damaging the graphene.

However, later studies found that, by analysing graphene's interfacial adhesive energy, it is possible to effectively separate graphene from the metallic board on which it is grown. Besides, it is also possible to reuse the board for future applications (theoretically, an infinite number of times), thereby reducing the toxic waste previously created by this process.

Further, quality of graphene that was separated by using this method was high enough to create molecular electronic devices successfully. While this research is very highly regarded, quality of graphene produced will still be the limiting factor in technological applications.

Graphene needs to be produced on very thin pieces of metal or other arbitrary surfaces (tens of nanometres thick) using chemical vapour deposition at low temperatures. And it has to be separated in a way that can control such impurities as ripples, doping levels and domain size, while also controlling the number and relative crystallographic orientation of graphene layers. With production techniques becoming more simplified and cost-effective, graphene will be more widely utilised in various applications in general and electronics in particular.

Graphene has a tremendous number of possible applications, and scientists and engineers are just getting started in their work to develop it. Let us hope that their sincere efforts in this direction will create good chances of developing new graphene based electronics applications that are faster, flexible, robust, cheaper and efficient. **EFY**